Hands-On Project 1-2 Exploring the Components of the Windows 10 OS

TIME REQUIRED: 20 minutes

OBJECTIVE: Explore Windows 10 components.

REQUIRED TOOLS AND EQUIPMENT: A Windows 10 computer or virtual machine, as specified in the "Before You Begin" section of the Preface

DESCRIPTION: In this project, you explore Windows 10 components by using some Windows system tools.

Step 1: Sign in to your Windows 10 computer with an administrator account.

Step 2:

You've just used one important OS component—the user interface. While everything you do on a computer demonstrates the progression of input, then processing, and then output, let's try some other obvious examples. For instance, right-click Start and click Windows PowerShell to start a PowerShell session. PowerShell is like a command prompt except that it's more powerful. Type 25*25 and press Enter. PowerShell returns the result of 625. PowerShell reads your input, processes it, and produces the output.

Step 3:

Try a more useful PowerShell function. Type Get-ComputerInfo and press Enter. (Capitalization is not important in PowerShell commands.) This time, the processing takes a little more time and the output is extensive. PowerShell tells you just about everything you want to know about your computer, including information about the processor, BIOS, and process. To see the output page by page, type Get-ComputerInfo | more and press Enter. The | more part of the command sends the output to a program called more that paginates the output. Press <Space> to proceed to the next page or q to quit. Type Get-Command and press Enter to see the long list of commands available in PowerShell. Close the PowerShell window.

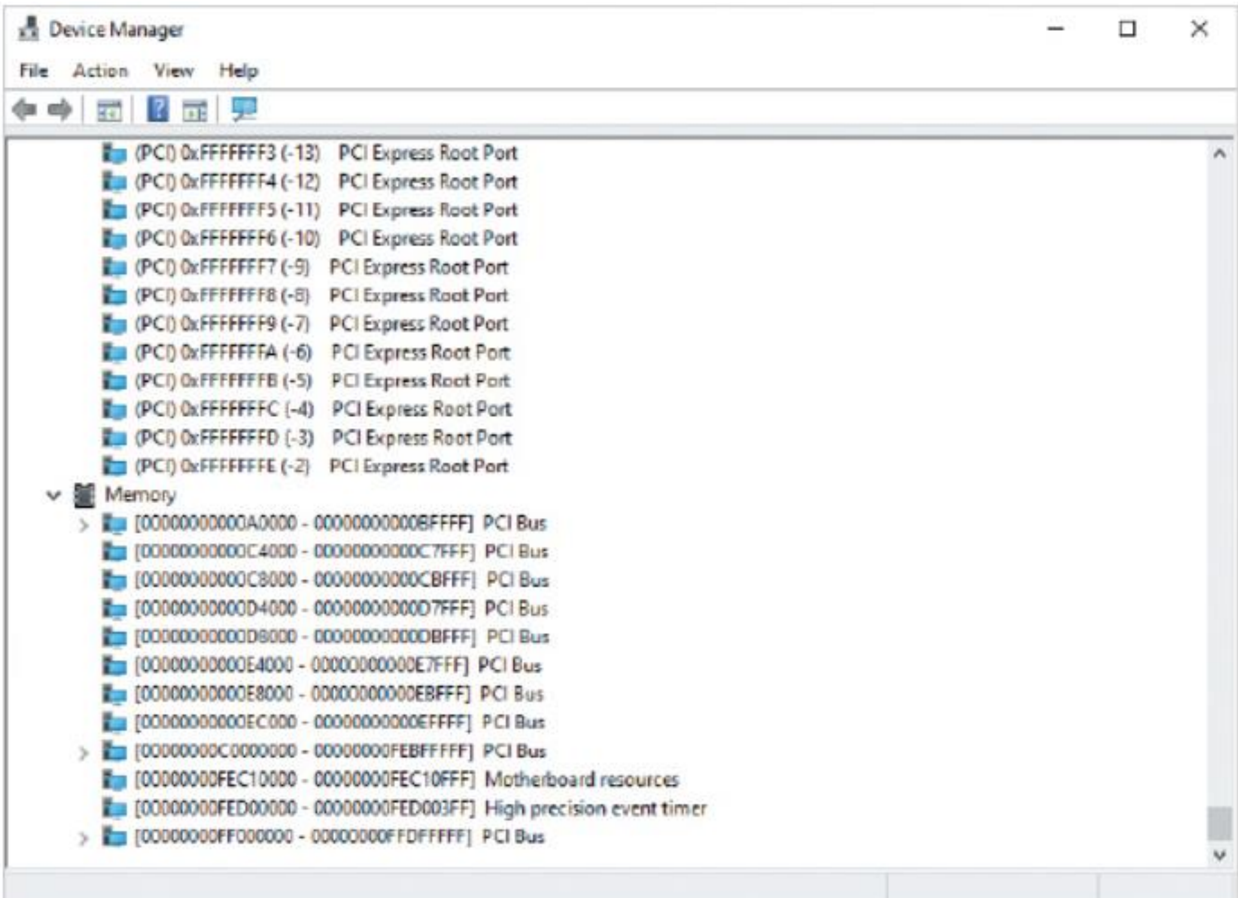
Step 4:

Right-click Start again to see a list of administrative tasks that are particularly important to an IT administrator or computer technician. Click System. The System command loads the Settings app, and you see a summary of your computer configuration in the right pane, including the processor, RAM, and Windows version. The left pane shows a list of settings categories. Close the Settings window.

Right-click Start and click Device Manager to see a list of I/O devices. Double-click Processors to see the type, speed, and number of processors installed.

Step 5:

Click the View menu and then click Resources by connection to see how the devices are using memory and interrupts. Double-click Interrupt request (IRQ) to see interrupt information, and then scroll down and double-click Memory to see how devices are using memory (see Figure 1-14). The details of what you see aren't important unless you have to troubleshoot a conflict or you are writing device drivers for Windows. Close Device Manager.



The Device Manager window shows a list of devices, including P C I Express Root Port activities, and under Memory, devices including P C I Bus, Motherboard Resources, and High precision Event timer.

Step 7:

Right-click Start and click Disk Management to see details about the installed storage on your computer. You'll work more with Disk Management in Module 4, "File Systems." Close Disk Management.

Step 8:

Right-click Start and click Computer Management. This tool contains a collection of other system tools, such as Device Manager and Disk Management, so you can manage much of the computer from a single tool. Click to expand Services and Applications and then click Services. You see a list of installed services and their status. Click the Status column heading twice to sort the services by status and bring the Running services to the top of the list. Close Computer Management.

Step 9:

Right-click the taskbar and click Task Manager. Click in the Type here to search box, type notepad, and press Enter. Move the Notepad window so you can see Task Manager. Click More details at the bottom of Task Manager. Under the Apps section, you see Notepad and Task Manager and then many processes under Background processes.

Step 10:

Right-click Notepad and click Go to details. The Details tab opens and notepad.exe is highlighted. You see its process ID (PID), status, amount of memory used, and other information. With notepad.exe highlighted, click End task. Click End process when prompted; the Notepad window closes and the process is no longer listed in Task Manager. Occasionally, you may have to use Task Manager in this way to terminate a hung process. Close Task Manager.

Hands-On Project 1-3 Working with Devices in Windows 10

TIME REQUIRED: 15 minutes

OBJECTIVE: Disable and enable a device.

REQUIRED TOOLS AND EQUIPMENT: Windows 10; a sound card with speakers or headphones attached is desirable

DESCRIPTION: In this project, you disable a device, reboot Windows, verify that the device is not functional, and re-enable the device.

Step 1: Sign in to your Windows 10 computer with an administrator account, if necessary.

Step 2: Right-click Start and click Device Manager.

Step 3:

Double-click Sound, video and game controllers. Right-click the device shown, take note of the options, and then click Disable device. Click Yes to confirm. If you are prompted to reboot, confirm and reboot the system. (Windows isn't entirely consistent on the issue of when a reboot is required.) Most devices allow you to update the driver, disable the device, and uninstall it. However, certain devices such as the keyboard, disk drives, and processor don't allow you to disable them.

Step 4:

If you had to reboot, sign in as an administrator. Notice on the far right side of the taskbar that the speaker icon in the notification area has a red circle with a white X, indicating the device is disabled. Right-click the speaker icon and click Open Sound settings.

Step 5:

Notice that the Output and Input sections indicate no devices are found. Keep the Settings window open, open Device Manager, and double-click Sound, video and game controllers, if necessary. Notice the down arrow on the audio device, indicating it is disabled. Right-click the audio device and click Enable device. You immediately see the Settings window reflect that a sound device is available and functioning. To verify, slide the Master volume slider to hear the Windows sound. Close the Settings window.

Step 6:

Sometimes a misbehaving device needs to be disabled and then re-enabled. Another option is to uninstall it and reinstall it. In Device Manager, double-click Network adapters, right-click the network adapter, and click Uninstall device. Click Uninstall when prompted.

Step 7:

The network adapter is gone in Device Manager, but of course it's still physically installed. Right-click the computer icon at the top of Device Manager; the icon bears the name of your computer. Click Scan for hardware changes. Windows looks for installed Plug and Play devices and installs the appropriate device driver if it can. In this case, it should find the network adapter and install the driver. Sometimes, uninstalling and reinstalling a device can solve a problem with the device because the driver is re-installed. If the problem remains, try rebooting the system after uninstalling the device. Close Device Manager.

Hands-On Project 3-1 Monitoring Processor Usage with Task Manager

TIME REQUIRED: 5 minutes

OBJECTIVE: Monitor processor usage.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: In this project, you use Windows Task Manager to monitor processor usage.

Step 1: Start your Windows 10 computer and sign in.

Step 2: Right-click the taskbar and click Task Manager on the shortcut menu. Click the More details button, if necessary.

Step 3: Click the Performance tab and click CPU in the left pane, if necessary (see figure below).

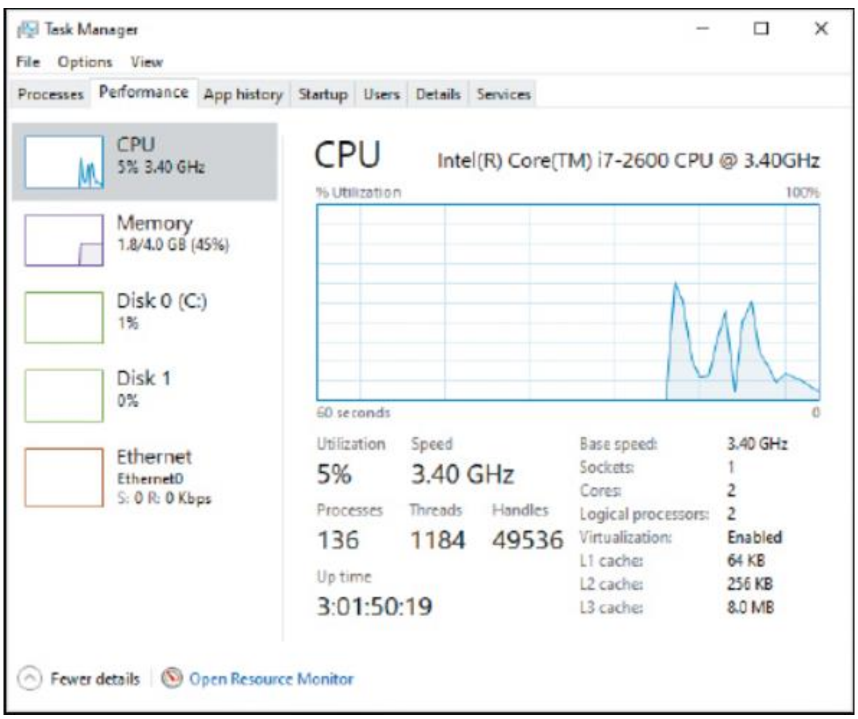


Figure 1: Task Manager with CPU Monitor

The Task Manager window with the Performance tab displayed. CPU is selected in the left pane. The CPU utilization information is shown in graph format. The Utilization is 5 percent. The speed is 3 point 40 gigahertz. The up time is 3 colon 0 1 colon 50 colon 19.

Step 4:

Watch the CPU history graph and note how it changes over time. Open and close a Web browser three or four times. Look at the CPU history graph now. You should see a distinct increase in CPU utilization.

Step 5:

Review the other information about the CPU. In Figure 3-6, you see the CPU model at the top of the graph. Below the graph, you see the maximum speed of the CPU, the number of cores, the number of logical processors, and the amount of L1, L2, and L3 cache, as well as other information. Your display will likely look different depending on the type of CPU running on your system. Can you tell by looking at Figure 1 if the CPU that is being monitored supports hyper-threading? How can you tell?

Step 6:

Click the Details tab. Click the CPU column to sort by the percentage of CPU time used by each process. Make sure the column values are in descending order so you see the largest numbers at the top. Most likely, you will see a process named System Idle Process at the top. This process is what the CPU executes when no other processes have work to do. A number close to 100 means that your computer is idle almost all the time.

Step 7:

Click Options, click Always on top, and then click the Details tab again. Open a program like a Web browser while watching Task Manager. You should see the System Idle Process with a lower CPU percentage briefly while the CPU reads from disk to load the Web browser into memory and then executes the Web browser program code.

Step 8:

Click the Performance tab and click Open Resource Monitor at the bottom of the window. Close Task Manager so you can see Resource Monitor. Click the CPU tab. In the left pane, you see processes that are loaded and how many threads each process has, plus other information. In the right pane, you see graphs showing total CPU usage, CPU usage by services, and then CPU usage for each logical processor (see Figure 2). If you have a two-core CPU with hyper-threading, you will see four logical processors. If you are running Windows in a virtual machine, you will likely see one or two logical processors depending on whether the processor supports hyper-threading.

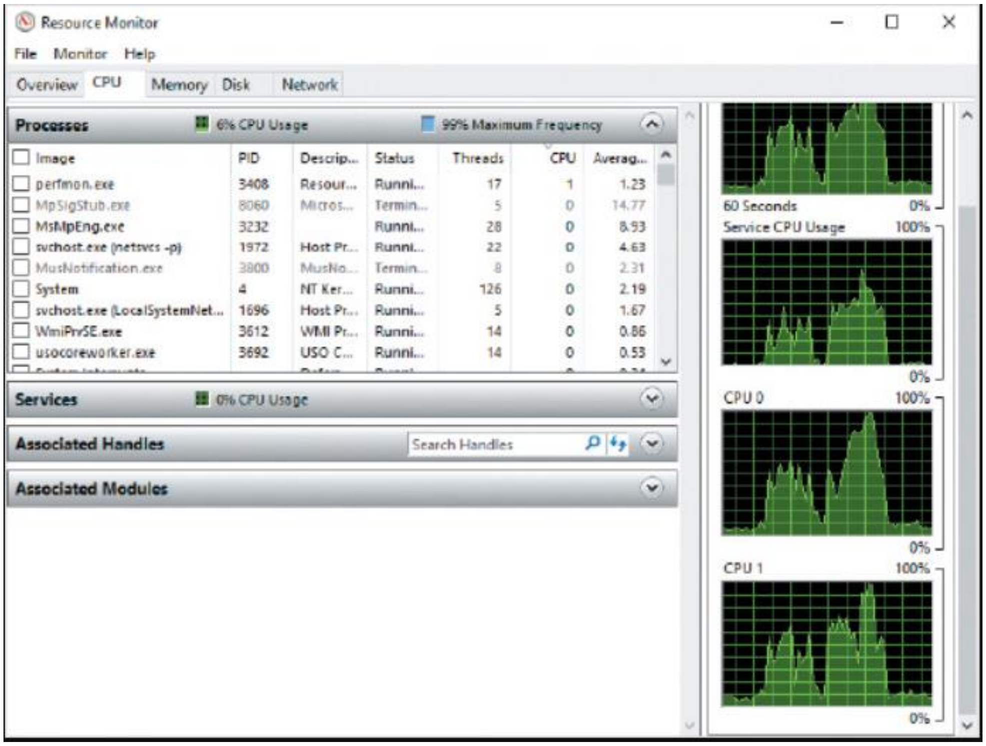


Figure 2: Resource Monitor

Hands-On Project 6-2 Using Device Manager in Windows 10

TIME REQUIRED: 10 minutes

OBJECTIVE: Use Device Manager to view driver information.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: In this project, you use Device Manager to view where to install or update a driver in Windows 10. You also use this utility to determine if a device is working properly and to view other information about the device.

Step 1: Log on to your Windows 10 computer, if necessary. Right-click Start and click Device Manager.

Step 2: Click to expand Display adapters.

Step 3: Double-click the specific adapter under Display adapters.

Step 4: Make sure that the General tab is displayed. You see the device status, which tells you the device is working properly.

Step 5:

Click the Driver tab (see Figure 3). Notice that you can click the Update Driver button to obtain an updated driver or install a driver if one is not already installed. Also, you can click the Roll Back Driver button to revert to a previously installed driver if there is a problem with an updated driver. (This button is disabled if you are installing the first driver.) You can click Disable to disable a device without actually installing it, and you can click the Uninstall button to remove a driver.

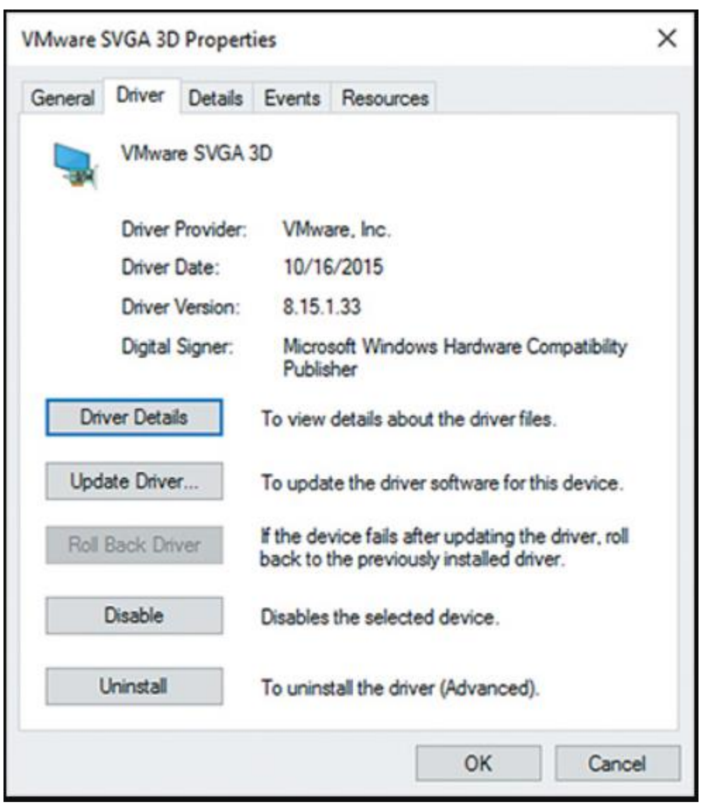


Figure 3: Video Adapter Information

Step 6: Click the Driver Details button.

Step 7: Display drivers usually have a number of associated files. The Driver File Details dialog box lets you view the name and location of the driver files. Click OK.

Step 8:

Click the Resources tab to see the resource settings for the display adapter. Notice there are one or more I/O ranges, one or more memory ranges, and an IRQ setting. A message at the bottom of the dialog box tells you if any conflicts are detected.

Step 9: Click Cancel and then close Device Manager.

Step 10: Stay logged on if you are continuing to the next project.

Hands-On Project 6-3 Configuring Mouse Settings in Windows 10

TIME REQUIRED: 10 minutes

OBJECTIVE: View mouse settings in Windows 10.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: In this project, you view the settings available for your mouse in Windows 10.

Step 1: Log on to your Windows 10 computer, if necessary.

Step 2: Click in the Search box, type mouse, and click Mouse settings in the results.

Step 3: Review the options available for changing the behavior of the mouse. Click Adjust mouse and cursor size.

Step 4: Try several of the size and color options to see how you can change the look of the mouse pointer and cursor.

Step 5: Click Additional mouse settings at the bottom of the screen to change the action of the buttons and scrolling. Click Additional mouse options to open the Mouse Properties dialog box.

Step 6: Click the Pointer Options tab and review the options. Click the Show location of pointer when I press the CTRL key check box.

Step 7: Press Ctrl. Notice the concentric circles displayed around the mouse pointer, which help you locate the mouse pointer easily. This option is very helpful when you are using multiple monitors.

Step 8: Click the remaining tabs to see the other mouse configuration options. Click Cancel when you are finished.

Step 9: Log off or shut down your computer.

Hands-On Project 8-1 Installing Hyper-V in Windows 10

TIME REQUIRED: 20 minutes

OBJECTIVE: Install Hyper-V in Windows 10.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: In this project, you install the Hyper-V feature on a Windows 10 computer.

Step 1: Start and log on to your Windows 10 computer.

Step 2:

In the search box, begin typing Windows Features until you see Turn Windows Features on or off in the search results, and click on it.

Step 3:

In the Windows Features window, click Hyper-V to select it. Click the plus sign to see the options under Hyper-V. Enable the management tools and the Hyper-V platform. Click OK.

Step 4: When prompted, click Restart now to finish the installation.

Step 5:

When Windows restarts, log on, click in the search box, and type hyper-v. In the search results, click Hyper-V Manager to open the Hyper-V Manager console.

Step 6:

In the next project, you will create a virtual machine and explore the settings. Leave Hyper-V Manager open if you are continuing to the next project.

Hands-On Project 8-2 Creating a VM in Hyper-V

TIME REQUIRED: 10 minutes

OBJECTIVE: Create a VM in Hyper-V in Windows 10.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: You have installed the Hyper-V role on Windows 10 and are ready to create a virtual machine.

Step 1:

Log on to your Windows 10 computer and open Hyper-V Manager, if necessary. If your Windows 10 computer is not selected in the left pane, click Connect to Server in the Actions pane, click Local computer, and then click OK.

Step 2: In the Actions pane, click New and then click Virtual Machine.

Step 3:

Read the information in the Before You Begin window. You can create a default virtual machine simply by clicking Finish in this window, but for this activity, click Next.

Step 4:

In the Name text box, type VMTest1. You can choose a location to store the virtual machine configuration, but for this activity, accept the default location of C:\ProgramData\Microsoft\Windows\Hyper-V by clicking Next.

Step 5:

In the Specify Generation window, you choose whether to create a generation 1 or generation 2 virtual machine. A generation 1 VM provides backward compatibility with older Hyper-V versions but provides fewer advanced features. Click Generation 2 and click Next.

Step 6:

In the Assign Memory window, type 1024 in the Startup memory text box, leave the option to use dynamic memory selected, and then click Next.

Step 7:

In the Configure Networking window, do not change the default option, Not Connected. Click Next.

Step 8:

In the Connect Virtual Hard Disk window, you can enter the virtual hard disk's name, size, and location. By default, the size is 127 GB, and Hyper-V assigns the hard disk the same name as the VM, with the extension .vhdx. You can also use an existing virtual disk or attach one later. Write down the location where Hyper-V stores the virtual hard disk by default, in case you want to access the virtual disk later. Click Next to accept the default settings.

Step 9:

In the Installation Options window, click Install an operating system later, if necessary. Click Next.

Step 10:

The Completing the New Virtual Machine Wizard window displays a summary of your virtual machine configuration. Click Finish. After the virtual machine is created, you return to Hyper-V Manager and see your new VM in the Virtual Machines pane of Hyper-V Manager.

Step 11: Leave Hyper-V Manager open if you’re continuing to the next activity.

Hands-On Project 8-4 Working with a Virtual Machine in Hyper-V Manager

TIME REQUIRED: 20 minutes

OBJECTIVE: Work with a virtual machine in Hyper-V Manager.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: You have installed a test VM that you can use to become familiar with managing virtual machines in Hyper-V Manager. In the following steps, you create a checkpoint, make some changes to the OS, and revert to the checkpoint.

Step 1:

Log on to your Windows 10 computer and open Hyper-V Manager, if necessary.

Step 2: Right-click VMTest1 and click Connect.

Step 3: Power on VMTest1 by clicking the Start toolbar icon.

Step 4:

After Windows Server 2019 boots, log on and open Notepad, and type your name in a new text document. Don't close Notepad or save the file yet. Click the Save toolbar icon or click Action and then Save from the menu of the Virtual Machine Connection console.

Step 5:

Close the Virtual Machine Connection console. In Hyper-V Manager, notice that the State column for the VM displays Saved or Saving. After the state has been saved, open the Virtual Machine Connection console by double-clicking VMTest1. Start the VM by clicking the Start toolbar icon. You're right where you left off in Notepad.

Step 6: Save the Notepad file to your desktop as file1.txt and then exit Notepad.

Step 7:

Click the Checkpoint toolbar icon or click Action and then Checkpoint from the menu. When prompted to enter a name for the checkpoint, type BeforeDeletingFile1 and then click Yes. In the status bar of the Virtual Machine Connection console, you see a progress bar that reads Taking checkpoint.

Step 8:

After the checkpoint is finished, minimize the VM and note that the checkpoint is listed in the Checkpoints section of Hyper-V Manager. Maximize the VM and delete the Notepad file you created. Empty the Recycle Bin to make sure the file is deleted.

Step 9: Click the Revert toolbar icon or click Action and then Revert from the menu.

Step 10

Click Revert when prompted. The VM displays a message that it’s reverting. When the desktop is displayed again, you should see the Notepad file back on the desktop. Close the Virtual Machine Connection console.

Step 11:

In Hyper-V Manager, right-click VMTest1 and click Shut Down. When prompted, click the Shut Down button. The Status column displays Shutting Down Virtual Machine.

Step 12:

After the VM state changes to Off, delete the checkpoint by right-clicking BeforeDeletingFile1 in the Checkpoints section and clicking Delete Checkpoint. Click Delete to confirm.

Step 13:

Close Hyper-V Manager, but stay logged on to Windows 10 if you are continuing to the next project.

Hands-On Project 8-3 Installing Windows Server 2019 in a VM

TIME REQUIRED: 30 minutes or longer

OBJECTIVE: Install Windows Server 2019 in a VM.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer and a Windows Server 2019 evaluation ISO file

DESCRIPTION: You have created a virtual machine and are ready to install Windows Server 2019 as a guest OS using the installation DVD. You need the installation media for Windows Server 2019.

You can download an ISO file for a trial version of Windows Server 2019 from www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2019. Be sure to download the ISO file, as you will have other options. Once the file is downloaded, you can attach the ISO file to the virtual DVD drive on your virtual machine by following the steps in this project.

Step 1: Log on to your Windows 10 computer and open Hyper-V Manager, if necessary.

Step 2:

Click VMTest1 in the Virtual Machines pane of Hyper-V Manager. In the Actions pane, click Settings under VMTest1 to open the Settings for VMTest1 window (see Figure 8-18).

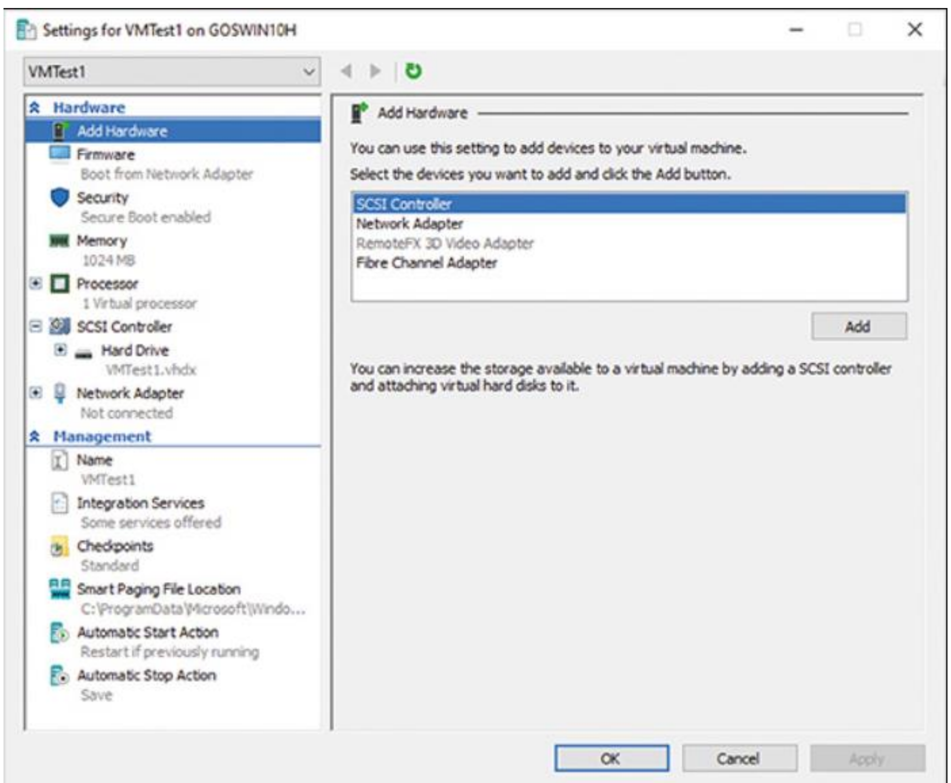


Figure 4: VM Setting in Hyper-V

Step 3:

Now you need to add a virtual DVD drive. Click SCSI Controller in the left pane. In the right pane, click DVD Drive and click Add. Click Image file and then click Browse. Locate and then click the Windows Server 2019 ISO file. Click Open. Click OK.

Step 4: In Hyper-V Manager, right-click the VMTest1 virtual machine and click Connect.

Step 5:

Power on VMTest1 by clicking Start. You will see a message to press any key to boot from the DVD. Be sure to click in the VM window and press the spacebar or any other key. If you are too late, just click Action and Reset in the Virtual Machine Connection console to restart the VM and try again.

Step 6:

Begin the installation of Windows Server 2019 by clicking Next in the Windows Setup window. From here, the installation steps are the same as those in Hands-On Project 5-4.

Step 7:

After Windows Server 2019 is installed, close the Virtual Machine Connection console and shut down the virtual machine by right-clicking VMTest1 and clicking Shut Down. Click Shut Down to confirm, if necessary.

Step 8: Leave Hyper-V Manager open if you’re continuing to the next activity.

Hands-On Project 9-2 Examining NIC Properties in Windows 10

TIME REQUIRED: 10 minutes

OBJECTIVE: Examine NIC properties.

REQUIRED TOOLS AND EQUIPMENT: Your Windows 10 computer

DESCRIPTION: When you describe a NIC as a component of networking, you're referring to both the hardware NIC and its driver. NIC drivers are configured in the OS in which they're installed and control certain operational aspects of the network interface as a whole. In this project, you examine the properties of your installed NIC. You also use the NIC's MAC address to look up the vendor. Not all NICs or NIC drivers have equivalent features, so your NIC might have more or fewer features than are described here.

Step 1: Start your Windows 10 computer and log on.

Step 2:

Right-click Start and click Network Connections. Click Change adapter options to open the Network Connections window. Right-click Ethernet0 and click Status. The Ethernet0 Status window shows a summary of information about your network connection. To see more information, click Details to open the Network Connection Details window (see Figure 5).

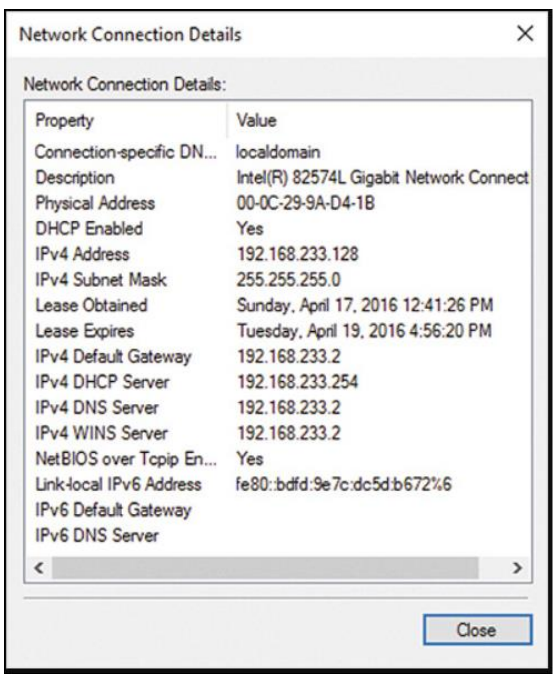


Figure 5: Network Connection Details

Step 3:

The Network Connection Details window shows information about your connection, including the NIC model, the physical (MAC) address, and your IP address configuration. Write down your MAC address, which you will use later to look up the NIC vendor. Review the remaining information and then click Close.

Step 4:

In the Ethernet0 Status window, click Properties. In the Ethernet0 Properties dialog box, click the Configure button under the “Connect using” text box. In the Network Connection Properties dialog box, click the Advanced tab (see Figure 6). Your NIC might have fewer, more, or different options.

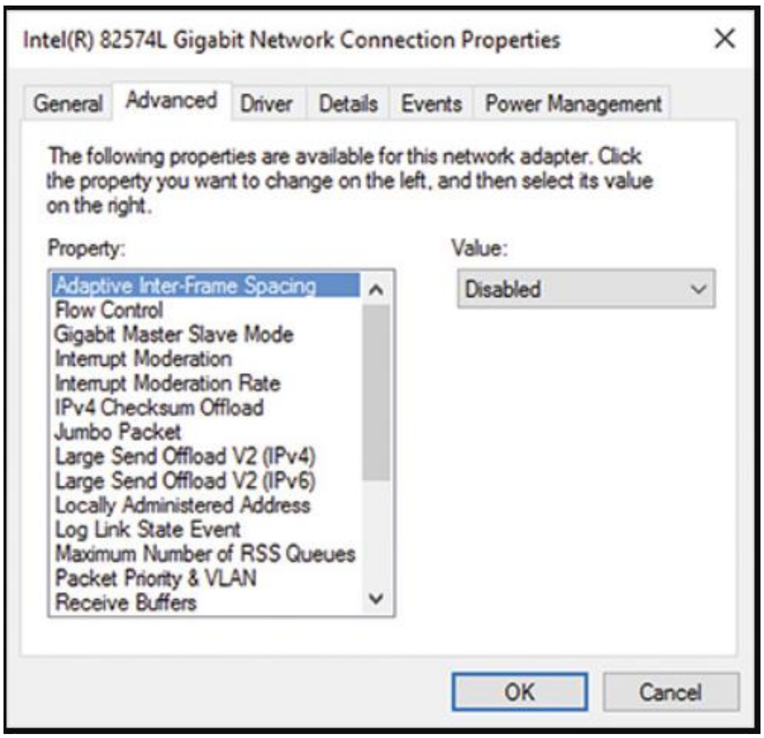


Figure 6: Connection Properties

Step 5:

Review the available properties for your NIC. When you select a property, you can see its possible values in the Value drop-down list.

Step 6:

Click Speed & Duplex (or Link Speed and Duplex) and then click the Value list arrow to see the possible values. On most NICs, the default value is Auto Negotiation, which means the NIC and switch exchange signals to determine the optimum operational mode. Other modes usually include combinations of 10, 100, and 1000 Mbps, full-duplex, and half-duplex. Normally, you don’t need to change these values unless auto-negotiation fails to work. If this happens, you’ll probably see the link status light change from on to off repeatedly or never turn on at all.

Step 7:

Click the Locally Administered Address property. (It might also be listed as Network Address, Physical Address, or MAC Address.) In most cases, this property's value is set to Not Present. You can use this property to override the NIC's burned-in MAC address by entering a new address in the Value text box. Normally, however, you shouldn't override the burned-in MAC address because if you duplicate an existing address accidentally, it can cause a loss of communication. Click Cancel to close the Network Connection Properties dialog box.

Step 8: Close the Ethernet0 Status window and the Network Connection Details window.

Step 9: Start a Web browser and go to www.coffer.com/mac_find.

Step 10:

In the "MAC Address or Vendor to look for" text box, type the first six digits of the MAC address you wrote down in Step 3. You don't need to enter the hyphen between each pair of digits, but you do need to enter any leading zeros. Click string to find the vendor of the MAC address. Knowing the vendor can help you track down devices that might be causing problems on your network.

Step 11: Close all open windows. Stay logged on if you're continuing to the next project.

Hands-On Project 11-1 Setting Password Policies in Windows

TIME REQUIRED: 10 minutes

OBJECTIVE: Set password policies in the Windows Local Security Policy console.

REQUIRED TOOLS AND EQUIPMENT: Windows 10

DESCRIPTION: This project shows you how to use the Local Security Policy console in Windows 10. You will set a password policy that specifies the following:

- Users must use 15 different passwords before re-using a password.
- Users must change their password every 90 days.
- Users can't change their password more often than every three days.
- The minimum password length is 12 characters.
- The password must contain three of these characteristics: uppercase letters, lowercase letters, numbers, and special characters.

Step 1: Sign in to your Windows 10 computer.

Step 2: In the search box, type secpol.msc and press Enter.

Step 3:

Click to expand Account Policies and click Password Policy. In the right pane, double-click Enforce password history. Set the value to 15 passwords remembered and then click OK.

Step 4: Double-click Maximum password age, set the value to 90 days, and click OK.

Step 5: Double-click Minimum password age, set the value to 3 days, and click OK.

Step 6: Double-click Minimum password length. Notice that by default, the minimum password length is 0, which means blank passwords are allowed. Set the value to 12 and click OK.

Step 7: Double-click Password must meet complexity requirements, click the Enabled option button, and click OK.

Step 8: Close the Local Security Policy console, but stay signed in if you are continuing to the next project.

Hands-On Project 11-2 Encrypting Files with EFS

TIME REQUIRED: 15 minutes

OBJECTIVE: Encrypt a file with EFS.

REQUIRED TOOLS AND EQUIPMENT: Windows 10

DESCRIPTION: This project shows you how to use EFS to encrypt a file.

Step 1: Sign in to your Windows 10 computer.

Step 2: Open File Explorer and click Local Disk (C:) in the left pane. Create a new folder named Private in the root of the C: drive.

Step 3:

Right-click the Private folder and click Properties. Click the Security tab to verify that the Authenticated Users group has at least Modify permission to the folder. When you test encryption, you want to see that another user can't access the encrypted file even if the account has the proper permission.

Step 4:

Click the General tab. Click the Advanced button. Click the Encrypt contents to secure data check box. By doing so, any file that you put in the Private folder is automatically encrypted. Click OK. Click OK again. You may see a prompt to back up your encryption key, which is a good idea, but for now you can ignore the message.

Step 5:

Double-click to open the Private folder and then right-click in the right pane, point to New, and click Text Document. Name the file passwords. You see a yellow padlock on the file icon, indicating the file is encrypted. Double-click the file, type any text you want in it, and then save and close the file.

Step 6:

Create another text document named nothingpersonal, enter some data in it, and save it. Next, you'll remove the encryption. Right-click the file and click Properties. On the General tab, click the Advanced button. Click to clear the Encrypt contents to secure data check box. Click OK. Click OK again.

Step 7:

The next step requires that you created a user named New Guest User 1 in Module 10. If you didn't create the user, do it now. The user password you create must meet the new complexity requirements and be at least 12 characters long.

Step 8:

Sign out of Windows and then sign in as New Guest User 1. If you created New Guest User 1 in Module 10, the password is Password01.

Step 9:

Open File Explorer and navigate to C:\Private. Double-click the passwords file you created in Step 5. Notepad opens, but you see a message indicating that you do not have permission to open the file. In reality, you have file permissions, but the encryption prevents you from opening the file. Click OK and close Notepad.

Step 10:

Double-click the nothingpersonal file and see that it opens in Notepad and you can change it and save it. You have file permissions to nothingpersonal, and because it is not encrypted, you can open it.

Step 11: Sign out of Windows.

Hands-On Project 11-3 Working with Windows Firewall

TIME REQUIRED: 15 minutes

OBJECTIVE: Explore and configure Windows Firewall.

REQUIRED TOOLS AND EQUIPMENT: Windows 10, Linux

DESCRIPTION: This project shows you how to use Windows Firewall. You need both your Linux computer and Windows 10 computer to be running at the same time so you can test the firewall settings. If you can't run both at the same time, you can work with a partner.

Step 1: Sign in to your Windows 10 computer and your Linux computer.

Step 2: On your Windows 10 computer, go to the search box, type firewall, and click Windows Defender Firewall in the search results.

Step 3:

In the left pane, click Turn Windows Defender Firewall on or off. In the "Customize settings for each type of network" window, you can turn the firewall on or off for the public and private network. If the firewall is on, you can block all incoming connections and select whether you should be notified when the firewall blocks a program. Click Cancel.

Step 4:

Click Restore defaults to return the firewall settings to their original status. Note that this will block file sharing even if you have shared folders. Click Restore defaults again and click Yes in the Restore Defaults Confirmation message box.

Step 5:

Open a command prompt or PowerShell window. Type ipconfig and press Enter to see your IP address. Make a note of your IP address.

Step 6:

On your Linux system, open a terminal window and type ping -c 3 ip-address , where ip-address is the IP address of your Windows 10 computer. The -c 3 part of the command tells Linux to send three ping packets. Without that parameter, Linux would continually try to send ping packets. The default firewall settings on Windows 10 should cause the ping to time out, although it may take a while. If you don't want to wait, press Ctrl+C to quit the ping program.

Step 7:

On your Windows 10 computer, click Advanced settings to open Windows Defender Firewall with Advanced Security. Notice the three profiles and note which one displays Active next to it. Click Inbound Rules.

Step 8:

Scroll down the Inbound rules pane until you see File and Printer Sharing (Echo Request - ICMPv4-In). This rule controls whether another computer can ping your machine; it is disabled by default. However, don’t configure that rule now.

Step 9:

Go back to the Windows Defender Firewall window and click Allow an app or feature through Windows Defender Firewall.

Step 10:

Find the File and Printer Sharing line and click the check box in the Private column (unless the active profile is public). Click OK.

Step 11:

Go back to the Windows Defender Firewall with Advanced Security window, click Action in the menu bar, and click Refresh to refresh the display. Scroll down until you see File and Printer Sharing (Echo Request - ICMPv4-In), and notice that the option now has a check box in a green circle next to it, indicating it is enabled. Several other File and Printer Sharing rules are enabled as well.

Step 12:

On your Linux system, try to ping your Windows 10 computer again. (You can press the Up arrow on the keyboard to repeat the last command.) The pings should be successful.

Step 13:

On your Windows 10 computer, double-click the File and Printer Sharing (Echo Request - ICMPv4-In) rule that is enabled. Click to clear Enabled and then click OK to disable the rule. Your pings will now be blocked but you can still share files.

Step 14: Try the ping again from the Linux computer to verify that it is blocked.

Step 15: Close all open windows, but leave your computers running for the next project.